Mahbod Tajdini

Master of Science in Data Science
ETH Zürich and UZH, Switzerland

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Portfolio | LinkedIn | GitHub

Education

Degree	Institute	CGPA	Year
MSc in Data Science	Universität Zürich (UZH) ETH Zürich (Special Student)	N/A (Ongoing)	2025 – Present
BSc in Artificial Intelligence	Vrije Universiteit Amsterdam (VU Amsterdam)	8.5/10 (=4.0 GPA)	2022 – 2025
Bachelor's Thesis: *Exploring Ball Recovery Times in Professional Football: Insights and Patterns*			

Research Experience

- **ETH Zürich - Computer Vision and Geometry Group (CVG) [🌐]** *Mar 2026 – Present*
  Graduate Student Researcher Zürich, Switzerland
  - Working on state-of-the-art dynamic SLAM under the supervision of Dr. Dániel Béla Baráth, integrating semantic scene graphs into deep visual SLAM by extending DROID-SLAM in the Wild (DROID-W) to improve efficiency and robustness in unconstrained environments

- **ETH Zürich - Ecosystem and Landscape Evolution Group (ELE) [🌐]** *Mar 2026 – Present*
  Research Assistant Zürich, Switzerland
  - Supporting the development of WildInSync, an initiative for global biodiversity monitoring that combines environmental DNA, remote sensing, and AI to generate standardized data for conservation research and decision-making

- **EPFL - Signal Processing Laboratory (LTS4) [🌐]** *Feb 2026 – Present*
  Visiting Student Researcher Lausanne, Switzerland
  - Conducting research on scalable cell-graph representations for digital pathology, focusing on batch-effect–robust graph construction, edge enhancement with tissue context, and graph diversity to improve generalization
  - Investigating multimodal foundation-model approaches that combine cell graphs and vision models, exploring efficient fine-tuning strategies and curated large-scale datasets for robust zero-shot and transfer learning

- **VU Amsterdam - Sports Intelligence Lab (SIL) [🌐]** *Jun 2025 – Mar 2026*
  Research Assistant Remote
  - Conducted data-driven research on ball recovery in professional football and explored machine learning applications in the sport under the supervision of Dr. Mauricio Verano Merino
  - Analyzed large-scale educational data to evaluate the effects of Generative AI tools on student outcomes in programming courses, contributing to a paper published at *ICSE 2026 (SEET Track)*

- **Forward Football [🌐]** *Sep 2024 – Jun 2025*
  AI Research Intern Amsterdam, Netherlands
  - Researched machine learning and computer vision models for football intelligence and tactical optimization
  - Analyzed large-scale spatiotemporal datasets to model player interactions, quantify team performance, and derive data-driven tactical insights

Work Experience

- **ETH Robotics Club (ETHRC) - Aeronautics [🌐]** *Apr 2025 – Present*
  Robotic Perception Engineer Zürich, Switzerland
  - Working with the Aeronautics team, where we use autonomous FPV racing drones to push the limits of high-speed aerial robotics, contributing to the perception stack by developing real-time visual-inertial odometry (VIO), gate detection, state estimation, Kalman filtering, sensor fusion, and motion capture to support environment perception, localization, and autonomous navigation in dynamic environments

- **ETH Zürich - Ecosystem and Landscape Evolution Group (ELE) [🌐]** *Jan 2026 – Present*
  Teaching Assistant Zürich, Switzerland
  - Serving as a Teaching Assistant for Environmental Systems Data Science: Machine Learning (701-3003-00L) under the supervision of Prof. Dr. Loïc Pellissier and Dr. Camille Pierre Albouy

- **Forward·Inc [🌐]** *Jan 2026 – Mar 2026*
  Student AI Consultant Amsterdam, Netherlands
  - Advising mission-driven organization on AI implementation through research and data analysis, evaluating AI tools and platforms, and delivering strategic roadmaps that align AI adoption with organizational goals and operational improvements

- **Universität Zürich - IFI [🌐]** *Sep 2025 – Dec 2025*
  Head Teaching Assistant Zürich, Switzerland
  - Served as Head Teaching Assistant (TA) for Informatics I (22AINF02) under the supervision of Prof. Dr. Harald Gall and Dr. Carol Alexandru, leading the TA team and managing teaching operations, exercise design, and assessment processes to ensure high-quality student learning support and effective course delivery

- **VU Amsterdam - Faculty of Science [🌐]** *Apr 2023 – Jun 2025*
  Teaching Assistant Amsterdam, Netherlands
  - Mentored new TAs through workshops, training sessions, and personalized feedback as a TA Mentor, enhancing instructional quality across the Faculty of Science
  - Taught and supported Machine Learning–focused courses and labs across 10+ undergraduate and Honors-level courses
  - Coordinated TA activities, contributed to course design, and facilitated communication between instructors and students as Head TA for the course: Computational Thinking (X_400475)

- **Baridsoft [🌐]** *Jun 2021 – Aug 2021*
  Information Technology Intern Tehran, Iran
  - Worked on computer systems, network infrastructure, and IT support, with hands-on experience in troubleshooting, device configuration, and maintaining internal systems

Research Publications

Capitalising on Football Data with Machine Learning: A Literature Review *Under Review - ACM Computing Surveys (ACM CSUR) 2026*
*Dionysios Kyriazopoulos, **Mahbod Tajdini**, Mauricio Verano Merino,*

The Clash of Codes: From Peer-to-Peer Duplication to AI-Generation in Introductory Programming Assignments *ICSE 2026 (SEET Track)*
Jose Maria Zuzarte Reis Claver, **Mahbod Tajdini***, Mauricio Verano Merino*

Programming Lego MINDSTORMS Robots using NI LabVIEW® *Open-access eBook - April 2022*
*Pooyan Nayyeri, **Mahbod Tajdini***

Positions of Responsibility

- **Hack4Good Organizing Staff,** Analytics Club at ETH Zürich *Nov 2025 – Present*
- **Executive Staff,** European Student Robotics Association (ESRA) *Nov 2025 – Mar 2026*
- **Voting member of the Program Committee for AI,** VU Amsterdam *Oct 2022 – Jun 2025*
- **Year Representative for BSc AI,** VU Amsterdam *Oct 2022 – Jun 2025*
- **International Student Ambassador,** VU Amsterdam *Oct 2022 – Jun 2025*
- **Robotics Competition Judge,** WRO Netherlands *Jun 2023 – Jul 2023*
- **Captain of the Student Robotics Team,** University of Tehran *Jun 2017 – Nov 2019*

Projects

- **ETH Zürich: ML in Finance [🌐][🐙]** *Aug 2025 – Oct 2025*
  - Evaluated the predictive efficacy of machine learning and deep learning architectures for Solvency II ratios under synthetic economic and stress scenarios
  - Implemented a reproducible SHAP-based interpretability framework to align algorithmic methodologies with actuarial principles, facilitating expert validation

- **Exploring Ball Recovery Times in Professional Football [🐙]** *Jan 2025 – Jun 2025*
  - Analyzed how key match events affect ball recovery times in elite football using open-access event data, and created game-timeline visualizations to reveal patterns around goals, substitutions, cards, and tactical shifts
  - Developed a reproducible Python pipeline to compute recovery metrics, perform statistical tests (normality, group differences, effect sizes), and present insights through interactive dashboards

Achievements

Academic Distinctions

- BSc in AI, VU Amsterdam – Graduated *cum laude* (Top 8% of cohort) *Jul 2025*
- IB Diploma Programme – Top 3% Worldwide *May 2022*

Hackathons

- *3rd place* at Datathon by Analytics Club at ETH – Switzerland's largest ML hackathon *Apr 2026*
- *4th place* at Anthropic Hackathon by Claude Builder Club of ETH Zürich *Mar 2026*

University of Waterloo - CEMC (Canadian Computing Competition)

- Awarded Certificate of Distinction as a *Top 25%* contestant worldwide *Dec 2020*
- Achieved *1st rank* in Iran *Dec 2020*

World Robot Olympiad (WRO)

- *1st place*, Football Category, Iran National Selection Competition *Oct 2019*
- *1st place*, Football Category, Iran National Selection Competition *Oct 2018*
- *16th place*, Football Category, Costa Rica International Finals *Dec 2017*
- *2nd place*, Football Category, Iran National Selection Competition *Oct 2017*

Skills

- **Programming Languages**: Python, C++, CUDA, R, SQL, Prolog
- **Machine Learning & Deep Learning**: PyTorch, PyTorch Geometric (PyG), TensorFlow, Scikit-Learn, HuggingFace, timm
- **Computer Vision & 3D**: OpenCV, Open3D, Kornia, scikit-image, lietorch, Gaussian Splatting, Ultralytics (YOLO), Stardist, OpenSlide, Roboflow
- **Robotics**: ROS 2, Gazebo, OpenVINS, PX4, Kalibr
- **MLOps & Experimentation**: Hydra, Weights & Biases, TensorBoard, Docker
- **Data Science & Scientific Computing**: NumPy, Pandas, SciPy, Statsmodels, Jupyter
- **Developer Tools**: Git, MySQL, LaTeX
- **Languages**: English (fluent), Persian (fluent), French (conversational), German (basic), Dutch (basic)

Certifications

- **Google**: Advanced Data Analytics; The Power of Statistics; Regression Analysis: Simplify Complex Data Relationships *Oct 2025*
- **DeepLearning.AI**: Generative Deep Learning with TensorFlow; Advanced Computer Vision with TensorFlow; Machine Learning *Sep 2025*
- **Stanford Online**: Advanced Learning Algorithms; Supervised Machine Learning: Regression and Classification; Unsupervised Learning, Recommenders, Reinforcement Learning *Aug 2025*
- **Anthropic**: AI Fluency Framework & Foundations; Model Context Protocol: Advanced Topics *Aug 2025*
- **IBM**: Tools for Data Science *Aug 2025*